THE FUNDAMENTAL EXACT SEQUENCE FOR A SMOOTH CURVE OVER AN ALGEBRAICALLY CLOSED FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument, written out as above; the closing sentence recalls the locally principal property of divisors, from which surjectivity follows immediately. That sentence is retained verbatim and this card does not expand it.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The standing set-up: a smooth curve X over an algebraically closed field with the inclusions of closed points and of the generic point. Same chapter, the set-up paragraph delivered verbatim as a context span in this card (source lines 10937-10941). Supplies the sheaves and morphisms appearing in the sequence.
- Structural sheaf, its units and the divisor description on a smooth curve. Same chapter together with the classical description of the exact sequence 1 -> Gamma(U, O_U^*) -> k(U)^* -> direct sum over points of Z. Supplies the well known exact sequence which the author identifies with the displayed sequence of sheaves.

Source reference. Original proposition 03PC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2592–2592. Navigation pointer invoked at proof line 10957; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

In this section, we fix an algebraically closed field k and a smooth curve X over k. We denote $i_x : x \hookrightarrow X$ the inclusion of a closed point of X and $j : \eta \hookrightarrow X$ the inclusion of the generic point. We also denote X_0 the set of closed points of X.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Theorem 0.1 (The Fundamental Exact Sequence)**.** *There is a short exact sequence of étale sheaves on X*

$$0 \longrightarrow \mathbf{G}_{m,X} \longrightarrow j_*\mathbf{G}_{m,\eta} \longrightarrow \bigoplus\nolimits_{x \in X_0} i_{x*}\underline{\mathbf{Z}} \longrightarrow 0.$$

Proof. Let $\varphi : U \to X$ be an étale morphism. Then by properties of étale morphisms (Proposition 03PC), $U = \coprod_i U_i$ where each U_i is a smooth curve mapping to X. The above sequence for U is a product of the corresponding sequences for each U_i, so it suffices to treat the case where U is connected, hence irreducible. In this case, there is a well known exact sequence

$$1 \longrightarrow \Gamma(U, \mathcal{O}_U^*) \longrightarrow k(U)^* \longrightarrow \bigoplus\nolimits_{y \in U_0} \mathbf{Z}_y.$$

This amounts to a sequence

$$0 \longrightarrow \Gamma(U, \mathcal{O}_U^*) \longrightarrow \Gamma(\eta \times_X U, \mathcal{O}^*_{\eta \times_X U}) \longrightarrow \bigoplus\nolimits_{x \in X_0} \Gamma(x \times_X U, \underline{\mathbf{Z}})$$

which, unfolding definitions, is nothing but a sequence

$$0 \longrightarrow \mathbf{G}_m(U) \longrightarrow j_*\mathbf{G}_{m,\eta}(U) \longrightarrow \left(\bigoplus\nolimits_{x \in X_0} i_{x*}\underline{\mathbf{Z}}\right)(U).$$

This defines the maps in the Fundamental Exact Sequence and shows it is exact except possibly at the last step. To see surjectivity, let us recall that if U is a nonsingular curve and D is a divisor on U, then there exists a Zariski open covering $\{U_j \to U\}$ of U such that $D|_{U_j} = \mathrm{div}(f_j)$ for some $f_j \in k(U)^*$. □